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RIS-LLM: Reasoning-informed semantic modeling of electricity market price dynamics☆

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ABSTRACT

Electricity prices are high-level indicators of modern energy system operations. Accurate modeling of market dynamics and electricity price forecasting (EPF) are therefore essential for the stability and efficiency of power markets. However, traditional models struggle to dynamically capture multiple-scenario influencing factors, leading to degraded predictive precision under high-volatility market conditions. To address this issue, this study proposes a Reasoning-Informed Semantic-Large Language Model (RIS-LLM) framework. It introduces a Reasoning Feature Extraction (RFE) module that dynamically identifies volatility-sensitive features through adaptive decomposition. By constructing a causal–volatility matrix, the RFE enables the model to reason across multiple driving factors, distinguishing transient fluctuations from structural trends. A Time Series Semantics (TSS) alignment mechanism is designed to semantically describe Granger-causal dependencies between environmental factors and electricity price dynamics. It employs a predictive-bias attention strategy to align volatility-driven features with interpretable semantic labels, such as demand surge or renewable oversupply, thereby highlighting the key determinants of electricity price fluctuations. Based on these, the RIS-LLM framework establishes a reasoning pathway from volatility features to semantic correlation structures. It enhances the model's ability to reason over multiple driving factors, enabling accurate forecasting of electricity price under dynamic market conditions. Experiments on a public benchmark dataset demonstrate that RIS-LLM achieves state-of-the-art forecasting accuracy. It delivers improvements in predictive accuracy while exhibiting stable performance even under highly volatile conditions.

1. Introduction

Modern energy systems are evolving into highly coupled and information-intensive infrastructures due to the large-scale integration of renewable energy and increasing demand-side variability [1]. Capturing the dynamic states of such systems poses significant challenges, as the relevant operational information is distributed across nonlinear and highly volatile data streams. In this context, electricity prices can be regarded as high-level indicators that implicitly encode the underlying operational and economic states of energy systems. Accurate electricity price forecasting (EPF) serves as the economic and societal cornerstone for the efficient and stable operation of modern electricity markets [2]. Precise forecasts are essential for generation companies and suppliers to formulate optimal bidding strategies and maximize profits. For large industrial consumers and energy traders, electricity price predictions serve as a crucial instrument for hedging financial risks and optimizing energy expenditures [3]. Meanwhile, grid operators rely on reliable forecasting to ensure system security, alleviate network congestion, and maintain the continuous reliability of power supply [4]. However, with the global energy transition, particularly the large-scale integration of variable renewable energy sources such as wind and solar, the operational paradigm of electricity markets is undergoing a fundamental transformation. The intermittency and uncertainty of renewables result in electricity prices exhibiting highly volatile and non-stationary behavior, diminishing the predictive power of historical price patterns [5]. Increasingly, price dynamics are driven by discrete, abrupt real-world factors, such as sudden weather

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Nomenclature

Abbreviations

ALOESS	Adaptive Locally Estimated Scatterplot Smoothing
ARIMA	Autoregressive Integrated Moving Average
ARMAX	Autoregressive Moving Average with Exogenous Variables
CNN	Convolutional Neural Network
CTSGAN	Conditional Time Series Generative Adversarial Network
EPF	Electricity Price Forecasting
FAEP	Feature-Augmented Electricity Price Prediction
FTS	Functional Time Series
GARCH	Generalized Autoregressive Conditional Heteroskedasticity
GRU	Gated Recurrent Unit
IDPs	Important Data Points
LLM	Large Language Model
LSSVM	Least Squares Support Vector Machine
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MASE	Mean Absolute Scaled Error
MIC	Maximum Information Coefficient
MSE	Mean Squared Error
PaP	Prompt-as-Prefix
RF	Random Forest
RFE	Reasoning Feature Extraction
RIS-LLM	Reasoning-Informed Semantic-Large Language Model
RMSE	Root Mean Squared Error
RNN	Recurrent Neural Network
RVM	Relevance Vector Machine
sMAPE	Symmetric Mean Absolute Percentage Error
SVM	Support Vector Machine
SVR	Support Vector Regression
TSS	Time Series Semantics
VCS	Virus Colony Search
ViT	Vision Transformer
VMD	Variational Mode Decomposition
WT	Wavelet Transform
XGBoost	Extreme Gradient Boosting

Symbols

C	Predictive dependency matrix
d_k	Dimension of the key vectors
E_i	Time-series embeddings
$\mathcal{F}$	Forecasting model
f_i	Fluctuation event
H	Forecasting horizon
h_0	Base smoothing parameter
$h_{a,t}$	Adaptive smoothing parameter
k	Sensitivity coefficient
$\mathcal{L}$	Forecasting loss
L_S	Semantic drivers
M_{attn}	Volatility-induced feature representations
M_{feat}	Feature matrix
N	Number of patches
P	Patch length
$\mathbf{P}$	Historical electricity price series
P_{proto}	Prototype embeddings
$\hat{p}_i$	Prototype
v_t	Volatility series
$\tilde{v}_t$	Normalized volatility series
W	Size of the sliding window
X	Input price series
$\mathbf{X}_t$	Vector of exogenous features
X_{fluc}	Fluctuation component
X_{trend}	Trend component

$\hat{Y}$	Future price sequence
$\mathbf{Y}_t$	True price sequence
Z	Hidden representation of predicted prototypes
γ	Significance level
λ	Causal guidance strength

changes, extreme climate events, unplanned outages of generation units or transmission lines, and geopolitical disturbances [6]. From an engineering informatics perspective, this volatility poses a significant challenge for the integration of variable renewable energy into the grid [7]. Accurate and interpretable forecasting is no longer merely a statistical requirement but a critical informatics foundation for net-zero energy design and real-time grid balancing [8]. By transforming raw price signals into actionable semantic knowledge, engineers can better manage the risks associated with the energy transition and optimize the deployment of storage systems.

To tackle the growing complexities in EPF, researchers have proposed various methodologies ranging from traditional statistical models and machine learning approaches to contemporary deep learning and Large Language Model (LLM) architectures. Despite their respective successes in capturing linear and non-linear temporal dependencies, existing methods are fundamentally constrained by a semantic gap between high-dimensional numerical time-series and the physical market cognitive space. They largely operate as numerical pattern recognizers, struggling to disentangle genuine causal drivers from multivariate noise or to dynamically adapt to out-of-distribution structural shifts caused by sudden real-world events.

To address the limited multi-scenario drivers and key factors analysis capability in existing forecasting approaches, this paper proposes a Reasoning-Informed Semantic Large Language Model (RIS-LLM) framework for EPF, with the following contributions:

- A Reasoning Feature Extraction (RFE) module is proposed to dynamically capture multi-scenario influencing factors. It performs volatility-aware adaptive decomposition to construct a causal–volatility matrix, enabling the extraction of volatility-sensitive features that distinguish transient fluctuations from structural trends.
- A Time Series Semantics (TSS) alignment mechanism is developed to semantically describe the Granger-causal dependencies between environmental factors and price dynamics. It employs a predictive-bias attention strategy to align volatility-driven features with semantic labels, highlighting the key determinants of electricity price fluctuations.
- This paper proposes a RIS-LLM framework that establishes a unified reasoning pathway from volatility features to semantic correlation structures, enabling a comprehensive understanding and accurate forecasting of electricity price under dynamic market conditions.

The remainder of this paper is organized as follows. Section 2 reviews the related works in EPF. Section 3 introduces the formulation of the EPF problem and provides a detailed description of the proposed RIS-LLM framework. Section 4 reports the experimental setup and presents the results together with comprehensive analyses. Section 5 concludes the paper and outlines potential directions for future research.

2. Related works

2.1. Statistical and traditional machine learning methods

Early EPF research concentrated on statistical time-series models. These approaches are valued for interpretability and ability to model linear temporal dependencies. Gonzalez et al. [9] proposed a Functional Time Series (FTS) forecasting model by extending the Seasonal Autoregressive Moving Average with Exogenous Variables (ARMAX) framework into the Hilbert space. The method introduces integral operators as functional parameters modeled with sigmoid functions, optimized by a Quasi-Newton algorithm. It effectively captures autoregressive, moving average, and exogenous functional terms. Empirical tests on the Spanish and German electricity markets showed better accuracy compared with reference models, reducing the prediction error by approximately 2% compared to the standard ARMAX model. However, a notable limitation of this approach is the challenge of determining the optimal number of sigmoid functions to prevent overfitting. Tan et al. [10] proposed a novel electricity price forecasting method by combining Wavelet Transform (WT) with Autoregressive Integrated Moving Average (ARIMA) and Generalized Autoregressive Conditional Heteroskedasticity (GARCH) models. The historical price series is decomposed into approximation and detail subseries via WT, and each subseries is predicted separately with suitable time series models. The final forecast is reconstructed from all subseries. Experiments on the Spanish and PJM electricity markets indicated that it reduces the average weekly mean error by 3.83% compared to a WT-neural network approach. However, a major limitation of this method lies in the empirical selection of the mother wavelet and decomposition levels. Rady et al. [11] applied a suite of tree-based methods—including decision trees and gradient boosted trees. There is also a classical ARIMA model to forecast commodity price time series. By fitting each model under a unified evaluation framework, the study compared their predictive ability using standard error metrics. The analysis highlighted the comparative strengths of ensemble tree methods over linear statistical baselines in capturing nonlinear temporal patterns in price dynamics. The results indicated that it outperforms decision trees and gradient boosted trees in predicting future gold prices, achieving the lowest Root Mean Square Error (RMSE) of 38.52. However, a significant limitation of utilizing tree-based algorithms like RF for time series forecasting is their inherent inability to extrapolate trends beyond the numerical range of the training data, making them vulnerable when forecasting unprecedented price highs or lows. Ray et al. [12] introduced a hybrid forecasting approach that integrates ARIMA for mean estimation and GARCH for volatility modeling. The framework leverages statistical modeling to capture linear and conditional variance structures. Comparative evaluations against standalone conventional statistical models, such as ARIMA, demonstrated the superiority of this hybrid architecture. Specifically, when applied to volatile agricultural price indices, the proposed model improved forecasting accuracy by 25% in RMSE, 28% in Mean Absolute Percentage Error (MAPE), and 29% in Mean Absolute Scaled Error (MASE). A potential limitation of such sequential two-stage hybrid models is the risk of error propagation, where inaccuracies in the initial linear ARIMA estimation strictly constrain the residual space. Despite their mathematical interpretability and adequate performance under stable market conditions, these traditional statistical methodologies share fundamental overarching limitations when applied to modern, highly volatile electricity markets. Primarily, they are inherently constrained by their strict reliance on linear assumptions, historical stationarity, and predefined functional forms. When the market experiences out-of-distribution structural breaks – such as unprecedented price spikes driven by complex external factors – these models suffer severe performance degradation due to their rigid mathematical boundaries. This fundamental inability to extrapolate beyond historical numerical ranges and capture complex, non-linear market dynamics highlighted a critical bottleneck in early EPF research.

To overcome the weak nonlinear modeling capability of statistical approaches, researchers introduced machine learning methods that enhance nonlinear feature learning, such as Support Vector Machines (SVM) [13] and Extreme Gradient Boosting (XGBoost) [14]. Albahli et al. [15] proposed using XGBoost for EPF to guide storage offloading in cloud data centers, aiming to reduce energy consumption costs. Empirical evaluations on the IESO dataset demonstrated that the XGBoost model achieves a predictive accuracy of 91%, outperforming the Random Forest (RF) and Support Vector Regression (SVR) benchmarks, enabling a 25.32% reduction in data center electricity costs. However, a major limitation of XGBoost in time-series forecasting is its lack of native temporal memory mechanisms; it relies heavily on manually engineered lag features and struggles to dynamically adapt to out-of-distribution market shocks without retraining. Zhang et al. [16] developed a forecasting framework based on SVM to tackle the nonlinear dynamics of electricity prices. The framework employs singular spectrum analysisto capture hidden patterns and multiple seasonalities, followed by a hybrid feature selection strategy using correlation thresholds and intelligent search algorithms, enhanced with Pearson, Spearman, and Kendall correlation measures. Validation on the New South Wales electricity market demonstrated the superiority of this approach, achieving a maximum reduction of 35.5% in RMSE compared to the standard SVM benchmark. But such SVM-based frameworks have high computational complexity when scaling to massive market datasets, coupled with their heavy reliance on external, manually designed feature extraction techniques. Mohsin et al. [17] proposed a deep convolutional neural network for forecasting crude oil prices using precious metal prices as predictors. The architecture incorporates a specialized convolution block and a regression layer to capture nonlinear relationships among multiple commodity markets. Benchmark comparisons demonstrated that the Convolutional Neural Network (CNN) effectively captures cross-market interactions, reducing the Mean Absolute Error (MAE) to 0.3245 compared to the best-performing decision tree baseline. However, the standard CNN architecture in sequential forecasting relies heavily on fixed-size convolutional kernels, which inherently limit its capacity to capture long-term temporal dependencies and dynamically adapt to sudden out-of-distribution market shocks. Ghasemi et al. [18] introduced a joint forecasting framework for price and load signals. The approach uses WT for denoising and mutual information for feature selection, followed by Least Squares Support Vector Machine (LSSVM) with GARCH to model linear and nonlinear patterns. A modified Virus Colony Search (VCS) algorithm optimizes LSSVM parameters. Tests on NSW electricity market validated the model's efficacy, achieving an exceptionally low average error variance of 0.0072 for weekly price forecasts. Despite these impressive quantitative metrics, the consecutive processes of wavelet denoising, feature selection, and heuristic optimization are highly prone to error accumulation across stages, which severely restricts the framework's real-time adaptability when encountering sudden structural market breaks. Lu et al. [19] developed a two-stage framework combining feature selection and neural forecasting for crude oil prices. Core influencing factors are first identified using elastic-net regularization, spike-and-slab lasso, and Bayesian model averaging. The selected variables are then used to train an Long Short-Term Memory (LSTM) network, with performance compared against various classical time-series and neural models. Results indicated that integrating rigorous variable selection with deep recurrent architectures strengthens both level prediction and directional accuracy. It outperformed the standard LSTM baseline by improving RMSE and direction by 1.13% and 11.63%, respectively. But a fundamental limitation of such disconnected two-stage frameworks is the static separation between feature selection and sequence modeling. The framework cannot dynamically realign its feature focus during sudden, out-of-distribution market shocks. Despite their enhanced capacity for non-linear modeling, these machine learning and hybrid frameworks are fundamentally constrained by their reliance on disjointed pipelines and manually engineered features. By treating feature extraction and predictive sequence modeling as separate, static stages, these models fail to learn temporal dependencies in an end-to-end manner. Consequently, they struggle to dynamically adjust their internal representations when exposed to sudden distributional shifts or complex, multi-scale market volatility. This inherent bottleneck in processing unstructured sequential data necessitated a paradigm shift towards continuous architectures capable of automated, end-to-end temporal representation learning.

2.2. Deep learning and Transformer architectures

With the rise of deep learning and Transformer architectures, models such as Recurrent Neural Network (RNN), LSTM, and Transformer-based frameworks have been increasingly adopted for EPF. These models can automatically extract temporal representations from unstructured price sequences, capturing long-term trends, local patterns, and short-range dependencies. Attention mechanisms further enhance forecasting by modeling multi-scale temporal relationships such as daily cycles, weather-driven variations, and high-frequency disturbances, while mitigating the impact of noise. Huang et al. [20] developed SEPNet, a hybrid short-term forecasting model integrating Variational Mode Decomposition (VMD), CNN, and Gated Recurrent Unit (GRU). VMD decomposes time series, CNN extracts features, and GRU learns dependencies. Their results demonstrated that SEPNet significantly outperformed various benchmark models, achieving an impressive four-season average MAPE of 0.73% and an RMSE of 0.453. However, the practical deployment of such a deeply ensembled architecture is often hindered by the complexity of the optimization process across all three distinct algorithmic modules. Wan et al. [21] proposed a CNN-LSTM-A model integrating CNN, LSTM, and attention mechanism. Pearson correlation selects features, CNN extracts high-dimensional features, LSTM models temporal correlations, and attention refines weight allocation. Results showed that the model achieves substantially lower forecasting errors, yielding a 7.3% improvement in prediction accuracy over the traditional LSTM baseline. While the attention mechanism successfully mitigates the degradation of long-term dependencies, the framework's reliance on linear Pearson correlation for initial feature screening may inadvertently discard non-linear variables that possess significant predictive value in complex power systems. Huang et al. [22] proposed a hybrid forecasting model combining Prophet and Transformer, enhanced by a Stacking strategy. Validated on four electricity market datasets, it demonstrated significantly improved accuracy and stability. The model recorded significant reductions in MAE, achieving 16.94 on the ENTSO-E dataset and 19.51 on the PJM dataset. However, the reliance on a stacked deep learning architecture inherently reduces the interpretability of the model, which can be a critical drawback for market regulators who require transparent pricing mechanisms. Yue et al. [23] presented a hybrid carbon-price forecasting model that integrates Hampel outlier detection, time-varying filtering empirical mode decomposition, and a transformer architecture. The decomposition stage isolates intrinsic mode functions to reduce non-stationarity, while a hyperparameter-tuned transformer performs multi-step and interval forecasting. Experimental validations across China's pilot carbon markets demonstrated exceptional predictive accuracy, with the model achieving average MAE of 2.26 for ten-step-ahead forecasts. While this heavily ensembled approach excels at quantifying uncertainty, the extensive preprocessing pipeline introduces significant computational latency, making it difficult to deploy in real-time, high-frequency trading environments. Yan et al. [24] proposed a multi-domain collaborative transfer learning, a missingness-aware forecasting framework that integrates multivariate singular spectrum analysis, multiscale covariate interaction, and multi-source collaborative transfer learning. Spectrum analysis reconstructs incomplete time series without explicit imputation, while the multiscale interaction module captures temporal dependencies. A collaborative transfer learning strategy further adapts a global model to site-specific data. Experiments demonstrated substantial performance gains, improving average forecasting accuracy by 10.5% under complete-data conditions and by 15.3% across various missing-data scenarios. But the frozen feature extraction layers in the method are unable to effectively capture the extreme local weather characteristics of the target site. Gezici et al. [25] explored transformer-based and patch-based architectures for financial asset forecasting by transforming time series into two-dimensional image representations enriched with technical indicators. Experimental comparisons showed that attention-based models, particularly Vision Transformer (ViT), more effectively exploited the spatial structures within these converted images compared to baseline convolutional architectures. Specifically, the proposed approach achieves superior performance in both predictive accuracy and broader financial evaluation metrics, especially when applied to frequently traded exchange-traded funds over extended holding periods. Although repurposing computer vision techniques yields promising directional forecasts, the prerequisite transformation of one-dimensional time series into two-dimensional pixel arrays inevitably disrupts the natural, high-frequency chronological continuity of the original market data. Mainstream deep learning and Transformer architectures have had remarkable success in modeling complex, multi-scale temporal dependencies, but remain limited by their purely statistical nature. They operate strictly as high-dimensional numerical pattern recognizers, failing to disentangle genuine causal physical drivers from spurious temporal correlations. Because these models process exogenous market variables without any comprehension of their underlying semantics, they are highly susceptible to feature entanglement and severe performance degradation during out-of-distribution structural shifts. This critical lack of explicit, scenario-based market reasoning establishes a definitive performance ceiling for purely data-driven algorithms, exposing the urgent need for a paradigm that can move beyond numerical curve-fitting to actual semantic deduction.

2.3. Large language models in time-series forecasting

The latest frontier explores using LLM for time-series analysis. Owing to their strong contextual reasoning capabilities, LLMs can integrate heterogeneous signals and learn higher-level semantics that are difficult for conventional models to capture. These abilities make LLMs a promising direction for improving robustness and scenario-aware forecasting. Jin et al. [26] introduced Time-LLM, a reprogramming framework adapting LLMs for time series forecasting. It reprograms inputs with text prototypes and applies Prompt-as-Prefix (PaP) for enhanced reasoning. Evaluations showed that Time-LLM achieved overall Mean Squared Error (MSE) reductions of 7.7% and 8.4% compared to GPT4TS in 10% and 5% few-shot learning settings, respectively. Furthermore, in zero-shot forecasting, the framework demonstrated a remarkable 22% error reduction against GPT4TS. Despite its remarkable few-shot capabilities, the standard Time-LLM framework directly patches raw time series without a dedicated feature refinement mechanism, leaving it susceptible to redundant noise in multidimensional data. Yu et al. [27] investigated the use of large language models for explainable financial forecasting by integrating historical prices, company metadata, and economic news. The study examined zero-shot and few-shot inference with GPT-4 as well as instruction-tuned Open-LLaMA models. Experiments of NASDAQ-100 stock return forecasting demonstrated that the proposed approach achieved significant performance gains, reducing the average stock price MSE by 10.9% for weekly predictions and 23.4% for monthly predictions when compared to the instruction-tuned OpenLLaMA-13B baseline. But treating continuous financial data simply as discrete text tokens restricts the model's ability to adaptively capture underlying non-stationary temporal structures. Wang et al. [28] developed MPP-GPT, a methanol price forecasting framework combining LLMs with machine learning. It employs Maximum Information Coefficient (MIC) for feature extraction and a Relevance Vector Machine (RVM) for residual correction. Empirical analysis on Chinese methanol prices demonstrated that this MPP-GPT architecture achieves exceptional accuracy, recording a MAE of 0.0996, a RMSE of 0.1083, and a MAPE of 3.96%. While the integration of MIC provides a degree of feature selection, its inherent pairwise nature evaluates indicators in isolation, potentially failing to eliminate complex multivariate noise or synergistic redundancies. Hu et al. [29] proposed a forecasting framework combining multiple attention mechanisms with LLMs to address challenges in cross-channel feature extraction and prompt construction. Through empirical validation, the results demonstrated substantial performance gains, reducing the MAE by 20.8% compared to standard LSTM networks, while maintaining a strong 16.6% MAE reduction even under zero-shot learning scenarios. Although eliminating prompt engineering and utilizing attention-based feature extraction enhances computational efficiency, relying exclusively on attention weights without a dedicated structural denoising mechanism may still leave the model vulnerable to extreme non-stationary fluctuations. Xue et al. [30] proposed Feature-Augmented Electricity Price Prediction (FAEP), a framework that incorporates LLM-driven feature extraction, external signals such as weather variables and volatility jumps, and a hybrid XGBoost-LSTM architecture. The use of retrieval-augmented generation enables more informative feature representations, while the downstream hybrid model refines temporal dependencies. Comparative evaluations suggested that the FAEP model significantly outperformed the competitive HARQ-L-CJ baseline, reducing the MAE and MSE by approximately 32.18% and 22.21%, respectively. Incorporating LLM-derived textual features and external variables extensively enriches the information space, but the absence of a rigorous feature refinement mechanism leaves the downstream hybrid model highly vulnerable to multivariate noise and feature redundancy. Lu et al. [31] proposed an agent-assisted forecasting framework that integrates LLMs with a Conditional Time Series Generative Adversarial Network (CTSGAN). Specifically, the architecture employs a fine-tuned LLaMa3 8B model as a bidding behavior agent and ChatGPT-4o as a market sentiment agent to generate multi-dimensional, high-resolution market covariates. These LLM-derived features are subsequently fed into a spike-enhanced CTSGAN to predict 5-minute resolution day-ahead electricity prices in the Australian National Electricity Market. Case studies indicated massive improvements over official system operator baselines, drastically reducing the RMSE from 147.92 to 47.28 and the MAE from 137.11 to 38.54. While deploying LLMs as external market agents effectively captures qualitative bidding and sentiment dynamics, the model relies entirely on the GAN to implicitly align these multi-agent inputs, which may struggle to maintain structural stability under the severe non-stationarity of 5-minute pricing. Existing LLM-based EPF methods have demonstrated profound potential by leveraging pre-trained knowledge and contextual reasoning capabilities to achieve high predictive accuracy. However, a critical, unifying limitation remains unresolved across these studies: the inherent semantic gap between high-dimensional numerical time-series and the LLM's linguistic cognitive space. As evidenced by the recent literature, directly patching raw numerical market data or relying solely on implicit neural alignments leaves these models highly vulnerable to multivariate noise and feature redundancy. More importantly, during the inference process, these unstructured architectures cannot guarantee that the LLM's attention is securely anchored to the truly causal physical drivers of electricity price fluctuations. Without a dedicated mechanism to extract and translate complex, multi-factorial numerical dynamics into structured semantic representations, current models struggle to execute reliable, scenario-aware market reasoning.

To synthesize the methodological evolution of EPF and further clarify the motivation of this study, Table 1 provides a comprehensive comparison of the key related works discussed above. As illustrated in the table, while existing approaches have progressively advanced the state-of-the-art in capturing temporal dependencies, they consistently exhibit structural or semantic limitations when confronted with highly volatile, out-of-distribution market conditions. This fundamental gap underscores the necessity of our proposed RIS-LLM framework, which aims to bridge high-dimensional numerical volatility with explicitly reasoned causal physical drivers.

Table 1
Summary of key related works and methodologies in electricity price forecasting.

Authors	Methodology	Dataset	Main Contributions	Limitations
Gonzalez et al. (2017) [9]	FTS	Spanish, German	Extended ARMAX to capture exogenous functional terms.	Prone to overfitting; strictly relies on linear assumptions.
Tan et al. (2010) [10]	WT-ARIMA-GARCH	Spanish, PJM	Decomposed subseries for separate statistical modeling.	Empirical selection of mother wavelet and decomposition levels.
Ray et al. (2023) [12]	ARIMA-GARCH	Agricultural price	Hybrid framework modeling mean and conditional variance.	Risk of error propagation from the initial linear estimation stage.
Albahli et al. (2020) [15]	XGBoost	IESO	Applied ensemble tree learning to reduce data center costs.	Lacks native temporal memory; struggles with out-of-distribution shocks.
Zhang et al. (2019) [16]	SVM	NSW	Captured hidden patterns via hybrid feature selection.	High computational complexity; relies on manual feature extraction.
Ghasemi et al. (2020) [18]	WT-LSSVM-GARCH	NSW	Joint forecasting for price and load signals.	Error accumulation across consecutive preprocessing and optimization stages.
Huang et al. (2021) [20]	SEPNet	ENGIE	Extracted multi-scale dependencies via deep ensembled modules.	High optimization complexity; operates purely as a numerical pattern recognizer.
Wan et al. (2023) [21]	CNN-LSTM-A	Steam turbine	Integrated attention to refine weight allocation.	Linear Pearson screening may discard non-linear variables.
Yue et al. (2023) [23]	EMD-Transformer	China carbon	Multi-step forecasting with outlier detection mechanism.	Extensive preprocessing pipeline introduces significant computational latency.
Jin et al. (2023) [26]	Time-LLM	ETT, ECL, ILI	Reprogrammed LLMs for zero-shot time-series forecasting.	Patches raw data without feature refinement; susceptible to multidimensional noise.
Wang et al. (2025) [28]	MPP-GPT	Chinese methanol	Combined LLMs with MIC and RVM for residual correction.	Pairwise MIC fails to eliminate complex multivariate synergistic noise.
Lu et al. (2025) [31]	CTSGAN	Australian NEM	Integrated multi-agent LLMs for market sentiment analysis.	Implicit GAN alignment struggles under severe market non-stationarity.

3. Methodology

3.1. Problem description

The EPF is a challenging task due to the highly nonlinear and volatile characteristics of electricity markets. Price fluctuations are not only determined by historical patterns but also driven by various exogenous factors [32]. Therefore, EPF can be modeled as a multivariate time series forecasting problem, where the objective is to predict future electricity prices.

Formally, let $\mathbf{P} = \{p_1, p_2, \ldots, p_T\} \in \mathbb{R}^T$ be the historical electricity price series up to time T. The goal of multi-step ahead forecasting is to predict the price sequence for the next H time steps. Given a look-back window of length L, let $\mathbf{X}_t \in \mathbb{R}^d$ represent the d-dimensional vector of exogenous features at time t, the model takes the historical data sequence to predict the future price sequence $\hat{Y} = \{\hat{\mathbf{P}}_{t+1}, \ldots, \hat{\mathbf{P}}_{t+H}\} \in \mathbb{R}^H$. The relationship between the past observations and future prices can be modeled as:

$$\{\hat{\mathbf{P}}_{t+1}, \ldots, \hat{\mathbf{P}}_{t+H}\} = \mathcal{F}(p_{t-L+1}, \ldots, p_t, \mathbf{X}_{t-L+1}, \ldots, \mathbf{X}_t; \Theta), \quad (1)$$

where $\mathcal{F}$ represents the forecasting model and Θ denotes its learnable parameters. The learning objective of the model is to minimize the discrepancy between the predicted and true prices over the forecasting horizon:

$$\Theta^* = \arg\min_{\Theta}; \mathbb{E}_t\left[\mathcal{L}(\hat{\mathbf{Y}}_t, \mathbf{Y}_t)\right], \quad (2)$$

where $\mathcal{L}(\cdot)$ denotes the forecasting loss, $\mathbf{Y}_t$ is the true price sequence, and the expectation is taken over all observed time windows.

The core challenge in EPF lies in the complex nature of $\mathcal{F}$. Electricity prices are not merely an autoregressive process; they are driven by a confluence of factors including demand-supply dynamics, fuel costs, weather conditions, and unforeseen real-world events. These drivers introduce strong non-linearities, non-stationarities, and abrupt structural breaks into the price series. Standard forecasting models often struggle to address these dynamics. Consequently, a robust model $\mathcal{F}$ must not only capture historical patterns but also understand the underlying causal structure and semantic context of price fluctuations.

3.2. Reasoning-informed semantic large language model (RIS-LLM) forecasting method

To address the aforementioned challenges, the RIS-LLM framework is proposed. As illustrated in Fig. 1, RIS-LLM is an end-to-end architecture designed to transform raw numerical time series into a semantically-rich representation that a LLM can effectively reason over. The framework sequentially processes the input data through three main modules: a RFE module that disentangles and characterizes the price series, a TSS alignment module that establishes a causal link between numerical features and their semantic drivers, and a core reasoning and forecasting module that leverages a reprogrammed LLM to predict future prices based on the enhanced semantic information.

3.2.1. Reasoning feature extraction (RFE)

The RFE module, depicted in Fig. 2, is designed to tackle the complex non-linear and volatile characteristics of electricity prices by decomposing the raw series into more interpretable components. This process involves two main steps: Adaptive Locally Estimated Scatterplot Smoothing (ALOESS) decomposition and Important Data Points (IDPs) based feature extraction.

The input price series $X \in \mathbb{R}^L$ is first decomposed into a trend component X_{trend} and a fluctuation component X_{fluc}. To address the heteroscedastic and volatile characteristics of electricity prices, an ALOESS

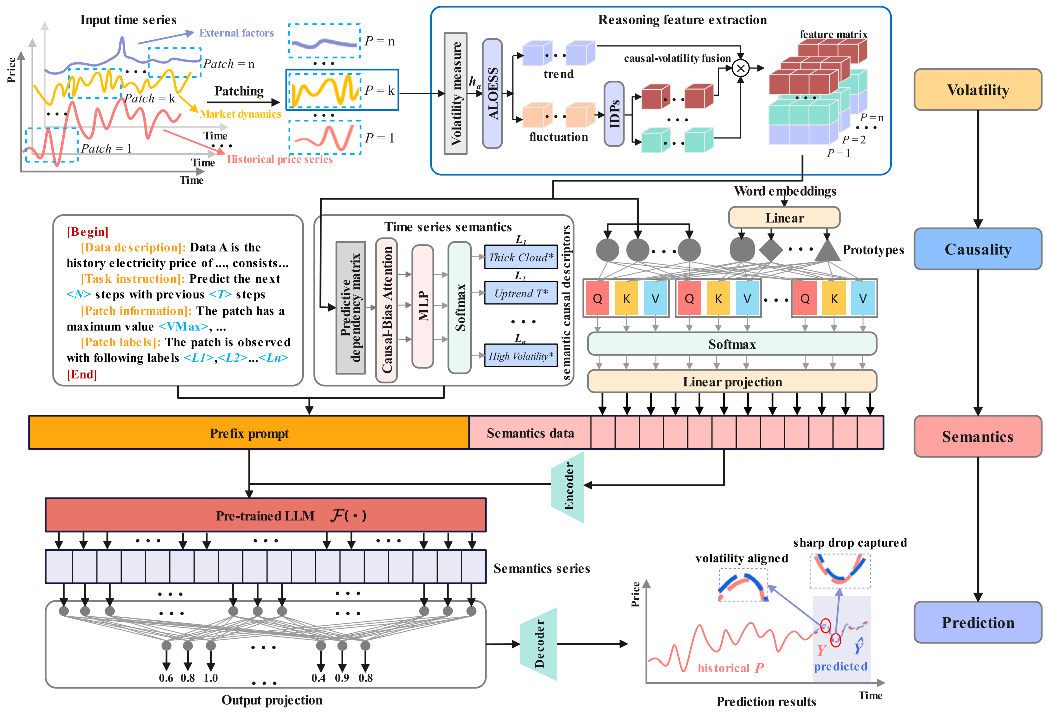

Fig. 1. The overall architecture of the proposed RIS-LLM framework.

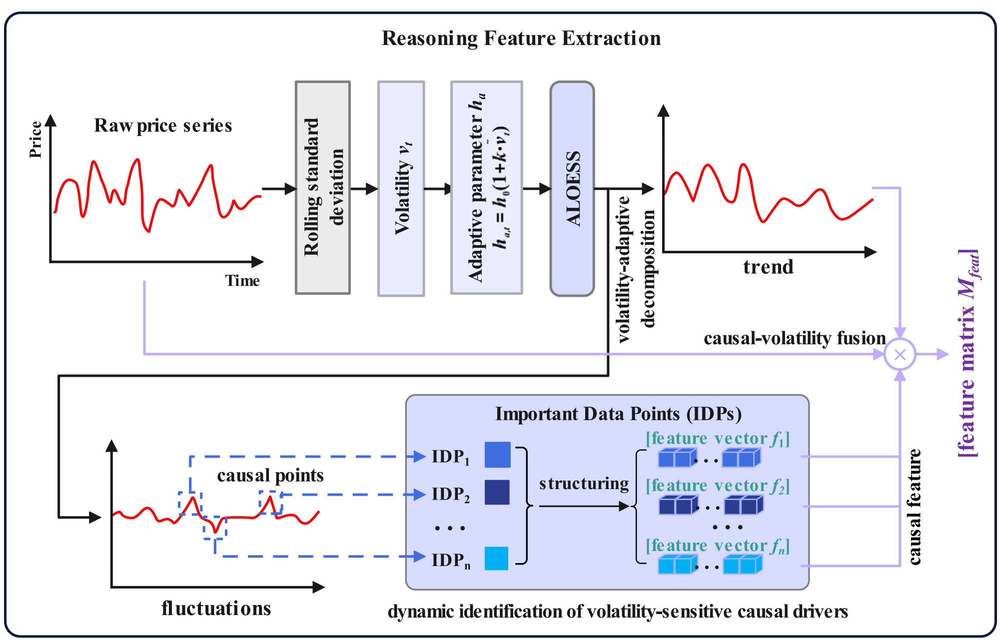

Fig. 2. The structure of reasoning feature extraction module.

method is introduced. In contrast to the conventional LOESS [33] that employs a fixed global smoothing parameter, the proposed approach dynamically adjusts the smoothing level at each time point according to the local volatility.

The process begins by quantifying the local volatility v_t at each time point t, calculated using the rolling standard deviation [34] over a sliding window:

$$v_t = \mathrm{RollingStdDev}(X, W), \tag{3}$$

where W means the size of the sliding window. The raw volatility values $\{v_t\}$ are then normalized to a range of $[0,1]$ to obtain the normalized volatility series $\{\tilde{v}_t\}$. Based on this, a point-wise adaptive smoothing parameter $h_{a,t}$ is defined for each time point t as a function of its normalized local volatility:

$$h_{a,t} = h_0(1 + k \cdot \tilde{v}_t), \tag{4}$$

where h_0 is a base smoothing parameter, and k is a sensitivity coefficient. At each time point t, the LOESS algorithm performs its locally-weighted regression using the specific local smoothing parameter $h_{a,t}$. This ensures that periods of high volatility are subjected to greater smoothing, causing the trend component to glide over transient spikes. Conversely, in stable periods, a smaller smoothing parameter allows the trend to follow the data more closely. This adaptive mechanism ensures that sharp fluctuations are not incorporated into the trend but are correctly isolated into the fluctuation component. The final decomposition is given by:

$$X = X_{trend} + X_{fluc}, \tag{5}$$

where X_{trend} is the output of the adaptive LOESS process. This provides a more accurate separation of long-term movements from short-term volatility.

The fluctuation component X_{fluc} contains crucial information about price spikes and sudden drops. To quantify these morphological characteristics, an IDPs [35] algorithm is employed. This algorithm first identifies all local extrema in the X_{fluc} series, which are regarded as "Important Data Points" since they indicate shifts in price volatility. For each pair of consecutive IDPs, a set of morphological features is extracted, including the magnitude of the price change, the duration between the points, the slope of the segment, and the area under the curve. This process transforms the volatile residual series into a structured set of feature vectors $\{f_1, f_2, \ldots, f_m\}$, where each f_i represents a distinct fluctuation event. Finally, the RFE module concatenates the trend component X_{trend} with the aggregated fluctuation features to form a comprehensive feature matrix $M_{feat} \in \mathbb{R}^{L\times D'}$, where D' is the dimension of the enriched features.

3.2.2. Time series semantics (TSS) alignment

The TSS module aims to establish an intrinsic informatics mapping between the multivariate features extracted by RFE and the underlying physical drivers of market volatility. This process functions as a knowledge representation layer that translates low-level numerical fluctuations into high-level engineering semantics. This is achieved through a causality-guided attention mechanism and an unsupervised semantic labeling process.

To move beyond simple correlation analysis, a pairwise Granger causality test [36] is performed across all feature dimensions in the matrix M_{feat}. The Granger causality test is a statistical method used to determine whether one time series provides useful information for forecasting another. For each feature pair (f_i, f_j), the null hypothesis that f_i does not Granger-cause f_j is tested. Based on the resulting p-values, a binary predictive dependency matrix $C \in 0,1^{D'\times D'}$ is constructed, where $C_{ij} = 1$ if the null hypothesis is rejected at a significance level of γ, and $C_{ij} = 0$ otherwise. This matrix C serves as a structured prior that encodes the directional predictive dependencies within the feature space.

Standard attention mechanisms can inadvertently focus on spurious correlations. To rectify this, the causal structure is incorporated into the attention mechanism. In the context of EPF, the query (Q), key (K), and value (V) matrices are learned latent representations obtained by linearly projecting the feature embedding M_{feat}. It encodes historical electricity prices and variables into three semantic spaces. Formally, these projections are defined as $Q = M_{feat}W_Q, K = M_{feat}W_K, V = M_{feat}W_V$, where W_Q, W_K, W_V are trainable transformation matrices. Conceptually, Q represents the information required at the current step to guide attention, such as the demand to identify which past volatility patterns or market signals are most relevant for forecasting the upcoming price movement. K encodes the semantic characteristics of historical time steps, including past price fluctuations, exogenous market conditions, and volatility-driven causal signals. V provides the corresponding content features from those historical steps—such as actual price levels, volatility patterns. The attention scores [37] are calculated as:

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d_k}} + \lambda C\right)V, \tag{6}$$

where d_k is the dimension of the key vectors, and λ is a learnable parameter that controls the strength of the causal guidance. By adding the scaled predictive dependency matrix C to the attention logits prior to the softmax operation, the model is explicitly encouraged to assign higher attention weights to feature pairs with established Granger-causal dependencies. This integration results in a more robust and interpretable feature representation M_{attn}.

To assign meaningful semantic labels to different categories of price fluctuations, an unsupervised clustering approach is adopted. The context vectors corresponding to significant fluctuation events are extracted from M_{attn}, after which the K-Means clustering algorithm is applied to partition these vectors into K distinct clusters. Each cluster represents a recurring morphological pattern of price volatility, such as sudden spikes, gradual increases, or oscillatory fluctuations. A high-level semantic label is subsequently assigned to each cluster, producing a set of semantic labels $L_S = L_1, \ldots, L_n$, where each L_i corresponds to a distinct type of price behavior. These labels offer a qualitative, human-interpretable summary of market dynamics, enabling the model to leverage semantic patterns for enhanced forecasting.

Finally, the causal attention maps are utilized to establish a latent correspondence between volatility-induced feature representations M_{attn} and their semantic drivers L_S. Through the end-to-end optimization of the RIS-LLM framework, this mapping enables the model to infer semantic meanings from causally-structured features, thereby achieving consistent alignment between causal features and their semantic interpretations.

3.2.3. RIS-LLM forecasting framework

The core of the proposed approach lies in the RIS-LLM framework, which reformulates the EPF task as a reasoning problem for a pre-trained LLM. Instead of fine-tuning the entire model, the LLM is reprogrammed through a carefully designed prefix prompt [26], while its internal parameters remain frozen. This paradigm allows the model to leverage the general reasoning and knowledge representation capabilities of the LLM without additional gradient-based training. The overall procedure of the proposed RIS-LLM forecasting framework is summarized in Algorithm 1.

Algorithm 1: RIS-LLM

Input: $X \in \mathbb{R}^L$, W, γ, P, H.
Output: $\hat{Y}$.

$v_t = \mathrm{RollingStdDev}(X, W)$;
$\tilde{v}_t \in [0, 1]$;
$h_{a,t} = h_0(1 + k\cdot\tilde{v}_t)$;
$X = X_{trend} + X_{fluc}$;
$M_{feat} \in \mathbb{R}^{L\times D'}$.
Perform pairwise Granger causality tests on M_{feat};
Set $C_{ij} = 1$ if $p_{ij} < \gamma$, else $C_{ij} = 0$;
$C \in \{0,1\}^{D'\times D'}$;
Obtain causally-aware feature representation M_{attn};
Derive semantic labels $L_S = \{L_1, \ldots, L_n\}$;
Establish latent mapping between M_{attn} and L_S.
Patching: $X = [X_1, X_2, \ldots, X_N]$, $\quad X_i \in \mathbb{R}^{P\times D}$;
Project patches into embedding space: $E_i = W_eX_i + b_e$;
For each E_i, select nearest prototype:

$$\hat{p}_i = \arg\max_{prot_j\in P_{proto}} \frac{E_i\cdot prot_j}{\|E_i\|\|prot_j\|};$$

Fuse E_i and $\hat{p}_i$ via cross-attention to obtain aligned embeddings;
Prompt = [DataDesc; TaskInst; PatchInfo; PatchLabels];
LLM performs reasoning to predict prototype indices for horizon H;
$\hat{Y} = W_oZ + b_o$;
return $\hat{Y}$

Following the patching strategy, the historical input sequence $X \in \mathbb{R}^{T\times D}$ is divided into several consecutive overlapped or non-overlapped patches as

$$X = [X_1, X_2, \dots, X_N], \quad X_i \in \mathbb{R}^{P\times D}, \tag{7}$$

where P denotes the patch length, and $N = \lfloor T/P \rfloor$ is the total number of patches. Each patch X_i is projected into the latent embedding space through a linear projection layer:

$$E_i = W_e X_i + b_e, \quad E_i \in \mathbb{R}^d, \tag{8}$$

where W_e and b_e denote the learnable projection parameters. These patch embeddings capture local temporal patterns and serve as the fundamental representation for subsequent reasoning.

To discretize the time-series representation space, a set of learnable prototype embeddings $P_{proto} = \{prot_1, prot_2, \dots, prot_M\}$ is introduced, where $M = 1000$ prototypes are adopted following the configuration in Time-LLM [26]. Each prototype corresponds to a semantic pattern derived from a linear projection of the vocabulary space. For each patch embedding E_i, the nearest prototype in the embedding space is selected based on cosine similarity:

$$\hat{p}_i = \arg \max_{prot_j \in P_{proto}} \frac{E_i \cdot prot_j}{\|E_i\|\|prot_j\|}. \tag{9}$$

The prototype sequence $\{\hat{p}_1, \dots, \hat{p}_N\}$ thereby represents the symbolic abstraction of the time series. To further enhance semantic alignment, a cross-attention layer fuses the original time-series embeddings $\{E_i\}$ with the corresponding prototype representations, ensuring consistent temporal–textual correspondence within the LLM embedding space.

The key to enabling the LLM's reasoning capability lies in the design of a comprehensive prefix prompt that integrates contextual, statistical, and semantic cues. The constructed prompt consists of four main components:

- Data Description (DataDesc): textual information describing the dataset domain and the statistical characteristics of its variables;
- Task Instruction (TaskInst): high-level natural language guidance specifying the prediction objective, horizon length, and evaluation focus;
- Patch Information (PatchInfo): quantitative descriptors of each patch, including statistical summaries such as mean, variance, and extrema;
- Patch Labels (PatchLabels): semantic annotations derived from the TSS module, providing interpretable labels that describe local temporal semantics.

All components are concatenated to form a structured prefix prompt:

$$\text{Prompt} = [\text{DataDesc}; \text{TaskInst}; \text{PatchInfo}; \text{PatchLabels}]. \tag{10}$$

The constructed prompt is prepended to the prototype sequence and fed into the frozen LLM. Leveraging its extensive pre-trained knowledge and in-context reasoning ability, the LLM processes this sequence to predict the prototype indices of future patches over the forecast horizon H. Finally, the predicted prototype embeddings are passed through an output projection layer to map them back into numerical values:

$$\hat{Y} = W_o Z + b_o, \tag{11}$$

where Z denotes the hidden representation of the predicted prototypes. This process transforms symbolic reasoning into continuous-valued forecasts, producing the final electricity price prediction $\hat{Y}$.

4. Experiments

4.1. Dataset

The electricity price forecasting dataset used in this study integrates generation and load data from the Spanish electricity market (ENTSO-E) [38], electricity price data sourced from ESIOS [39], and weather data for Madrid obtained via the Open Weather API [40]. The dataset spans the period from January 1, 2021, to December 31, 2025, with an hourly resolution. It contains 13 features, categorized as follows:

- Weather features: temperature, humidity, pressure, wind speed, wind direction, precipitation, and cloud cover;
- Generation features: fossil generation, renewable generation, and other generation;
- Consumption feature: total electricity demand;
- Price features: day-ahead electricity price and real-time electricity price.

Fig. 3 illustrates the real-time electricity price in January 2024, where it can be observed that both weekdays and weekends exhibit irregular fluctuations. At the weekly scale, the price series does not display clear periodicity. This observation is further supported by the daily heatmap in Fig. 4(a), where the absence of consistent horizontal banding suggests that price intensities vary stochastically rather than following a repetitive weekly pattern. Furthermore, the lag-168 scatter plot in Fig. 4(b) reveals a dispersed cloud structure rather than a linear alignment along the diagonal. This shows the weak correlation between weekly data of dataset. These results are consistent with the previously described characteristics of electricity price data, namely its high volatility and nonlinearity.

4.2. Baselines and evaluation metrics

To evaluate the forecasting performance of the proposed method, five representative time series prediction models were selected as baselines: the traditional machine learning method XGBoost [41], the deep learning method CNN-LSTM [42], the transformer-based method Autoformer [43], the trend decomposition-based method DLinear [44], and the recent large language model-based approach Time-LLM [26].

All baseline methods were evaluated using three widely adopted metrics: MAE, MSE, and Symmetric Mean Absolute Percentage Error (sMAPE). Their mathematical definitions are given as: $\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i|$, $\text{MSE} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$, $\text{sMAPE} = \frac{100\%}{N}\sum_{i=1}^{N}\frac{2|y_i-\hat{y}_i|}{|y_i|+|\hat{y}_i|}$, where y_i denotes the actual electricity price at time step i, $\hat{y}_i$ represents the predicted electricity price, and N denotes the forecasting length. MAE directly measures the average absolute deviation between predictions and ground truth. MSE penalizes larger errors more heavily due to the squared term, while sMAPE offers an intuitive percentage-based interpretation of forecasting accuracy.

4.3. Results

All experiments were conducted on a single NVIDIA RTX 4090 GPU with 48 GB memory. For both Time-LLM and the proposed RIS-LLM, the backbone large language model is LLaMA-7B, loaded with 4-bit quantization. This is a choice based on the balance between reasoning ability and computational latency. To comprehensively evaluate the model's generalization capabilities across distinct seasonal modalities and avoid selection bias, we adopted a rigorous chronological data splitting strategy over the dataset. Specifically, the final full calendar year (January 1, 2025, to December 31, 2025) was exclusively reserved as the test set to ensure a robust, full-cycle assessment. The remaining four years of historical data (2021–2024) were partitioned into training and validation sets at a 7:3 ratio. Training was performed for 10 epochs with a learning rate of 0.01. Other hyperparameters of the proposed method follow the settings of Time-LLM to ensure fair comparison. The complete implementation of RIS-LLM is publicly available at the GitHub repository: https://github.com/Huang-Mg/RIS-LLM.

The forecasting performance of all methods across different prediction horizons (12, 24, 48, and 72 h) is summarized in Table 2. Six representative approaches were compared: XGBoost, CNN-LSTM, Autoformer, DLinear, Time-LLM, and the proposed RIS-LLM. The results show that RIS-LLM consistently achieves the lowest MAE and MSE values across all horizons, highlighting its superior capability in capturing both short-term volatility and long-term trends. Notably, DLinear achieves the lowest sMAPE at 24 h horizon, which can be

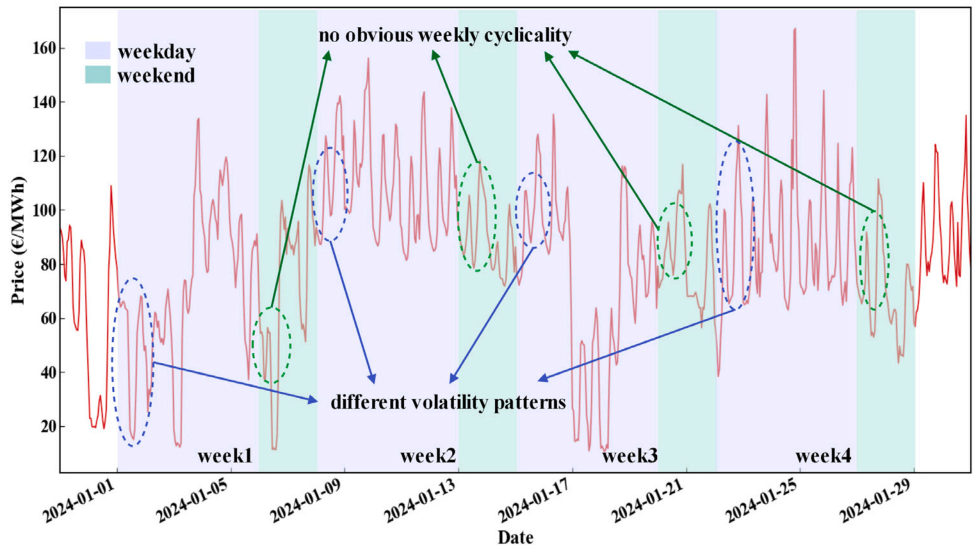

Fig. 3. The real-time electricity price in the dataset.

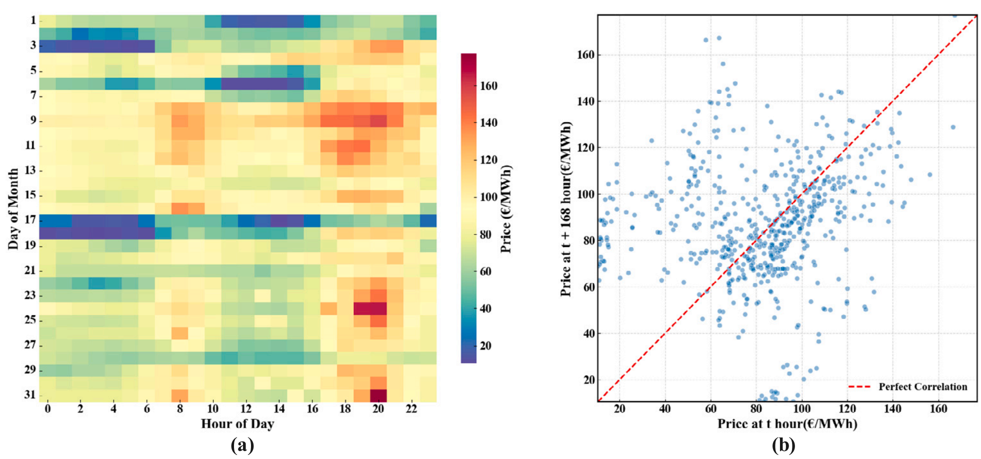

Fig. 4. The cyclicality analysis of hourly prices: (a) daily heatmap analysis; (b) lag scatter analysis.

attributed to its strong linear fitting ability that aligns well with local fluctuations in the short-term. However, as the horizon extends to 48 h and 72 h, the performance of DLinear degrades significantly, while RIS-LLM maintains stable improvements, demonstrating its robustness in long-range forecasting.

To evaluate the effectiveness of the adaptive smoothing parameter h_a in the RFE module, experiments are conducted with different fixed smoothing values and compared them with the adaptive strategy. The results are illustrated in Fig. 5. When a small smoothing parameter is applied, the trend decomposition module tends to absorb sudden price spikes into the trend component, resulting in a loss of critical fluctuation signals. Conversely, when a large smoothing parameter is used, the module over-smooths the curve, removing meaningful trend variations. In contrast, the adaptive smoothing parameter h_a dynamically adjusts based on local volatility, ensuring a balanced decomposition that preserves essential trend information while effectively extracting fluctuation signals. This validates the necessity of incorporating volatility-driven adaptivity in the decomposition step.

In the TSS module, semantic labels are generated through an unsupervised clustering process applied to high-dimensional volatility features. To objectively validate that these generated labels represent genuine, interpretable physical market drivers rather than mere mathematical artifacts, we conducted a rigorous post-hoc statistical analysis. We extracted the actual physical values of key exogenous variables: actual total load, renewable generation, and fossil generation. Table 3 presents the statistical validation results, including the mean and standard deviation for each category. A Kruskal–Wallis non-parametric test was employed to examine the variance across groups. The results yield a p-value of < 0.001 across all variables, confirming that the physical distributions of exogenous factors differ significantly across the generated semantic categories. To further visualize these dynamics, Fig. 6 illustrates the actual distributions of the exogenous variables under different semantic labels. The correlations are highly consistent with physical market logic: the cluster labeled as "Demand Surge" corresponds to a significantly elevated median grid load; the "Renewable Oversupply" label strictly aligns with exceptionally high generation renewable outputs; and the "Fossil Oversupply" category matches peaks in fossil fuel generation. This quantitative and visual evidence robustly demonstrates that the proposed TSS module accurately captures and semantically translates real-world physical mechanisms into actionable market intelligence.

To further evaluate the impact of the causal attention mechanism, feature attention maps are visualized with and without the incorporation of the predictive dependency matrix, as shown in Fig. 7. In the absence of causal constraints (Fig. 7a), the attention network distributes

Table 2
Forecasting performance of all methods.

Horizon		12	24	48	72
XGBOOST [41]	**MAE**	0.718	0.765	0.814	0.827
	MSE	0.921	0.968	1.201	1.294
	sMAPE	1.104	1.032	1.067	1.124
CNN-LSTM [42]	**MAE**	0.582	0.596	0.611	0.728
	MSE	0.651	0.720	0.747	0.958
	sMAPE	0.694	0.732	0.907	0.984
Autoformer [43]	**MAE**	0.604	0.607	0.687	0.701
	MSE	0.788	0.790	0.801	0.835
	sMAPE	0.887	0.895	1.006	1.029
DLinear [44]	**MAE**	0.518	0.567	0.634	0.658
	MSE	0.592	0.652	0.683	0.728
	sMAPE	0.674	**0.696**	0.861	0.940
Time-LLM [26]	**MAE**	0.501	0.543	0.598	0.625
	MSE	0.593	0.632	0.691	0.720
	MAPE	0.692	0.752	0.850	0.905
RIS-LLM	**MAE**	**0.474**	**0.509**	**0.571**	**0.599**
	MSE	**0.537**	**0.587**	**0.659**	**0.705**
	sMAPE	**0.629**	0.704	**0.804**	**0.839**

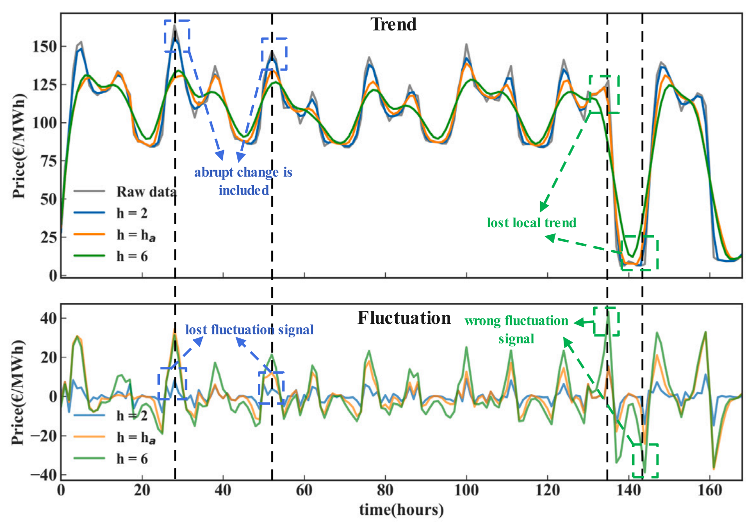

Fig. 5. Experimental results of the influence of adaptive parameter h_a on the trend decomposition effect.

Table 3
Statistical validation results of semantic labels.

Exogenous variable	Demand surge	Renewable oversupply	Fossil oversupply	Kruskal–Wallis p-value
Total load actual	**123376 ± 11026**	103832 ± 14169	102439 ± 17209	<0.001
Generation renewable	48669 ± 19606	**71600 ± 20438**	23867 ± 12111	<0.001
Generation fossil	23940 ± 7378	15073 ± 4354	**30613 ± 11726**	<0.001

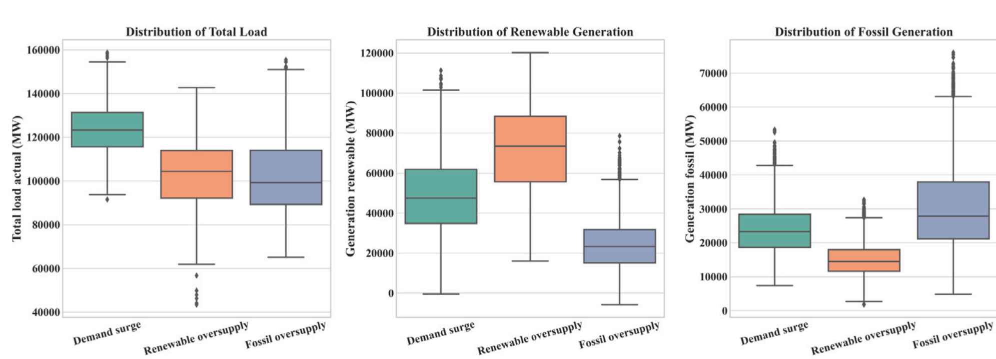

Fig. 6. Actual distribution of key exogenous variables under different semantic labels.

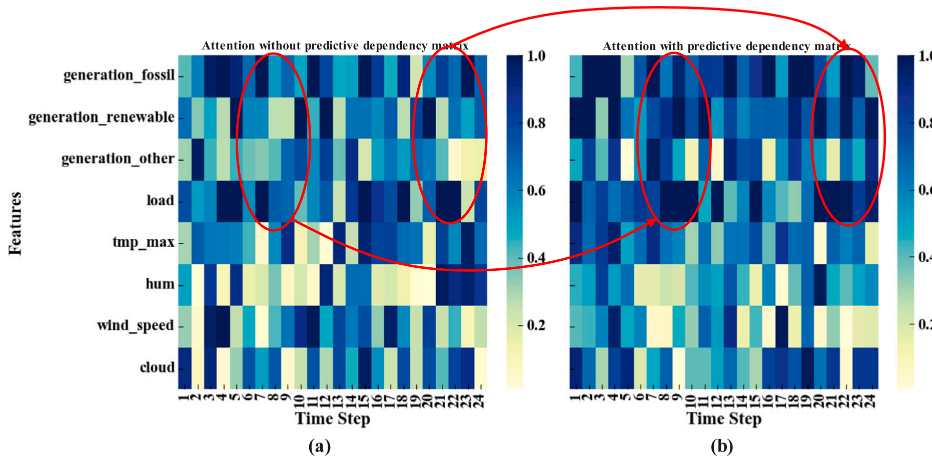

Fig. 7. Effect of predictive dependency matrix on attention network: (a) attention without predictive dependency matrix; (b) attention with predictive dependency matrix.

weights more uniformly across features, which may amplify noise. Upon introducing the predictive dependency matrix (Fig. 7b), higher attention weights are assigned to features exhibiting stronger causal relevance to electricity price fluctuations, such as fossil generation, renewable generation, and load demand. This visualization confirms that the predictive dependency matrix effectively directs the attention mechanism towards meaningful dependencies.

A primary motivation of this study is to overcome the predictive bottlenecks caused by extreme price spikes and high volatility in modern electricity grids driven by renewable energy integration. To explicitly evaluate the robustness of the proposed framework against these extreme market dynamics, a dedicated comparative analysis was conducted on isolated testing periods characterized by severe consecutive fluctuations and sharp price spikes.

Fig. 8 illustrates the predictive trajectories of all evaluated models during a representative high-volatility window. As depicted, the traditional machine learning baseline, XGBoost, generates extensive spurious volatility, failing to maintain structural stability. Meanwhile, standard sequence modeling architectures-including Autoformer, CNN-LSTM, and DLinear-exhibit a pronounced smoothing effect; they manage to fit the general trend but fundamentally fail to capture the high-frequency consecutive fluctuations. Time-LLM struggles at the absolute extreme values, explicitly missing the primary volatility spikes. In stark contrast, the proposed RIS-LLM demonstrates exceptional resilience. By leveraging reasoning-informed semantic alignment, it successfully captures both the sudden price spikes and the intricate sequential patterns of consecutive market fluctuations, maintaining a trajectory highly consistent with the ground truth.

Table 4 quantitatively corroborates these visual observations. During these extreme market conditions, RIS-LLM achieves the lowest prediction errors, recording an MAE of 0.273 and an MSE of 0.307. Compared to the best-performing baseline in this scenario, RIS-LLM significantly reduces the Mean Absolute Error by 25.2%. This dedicated analysis conclusively demonstrates that extracting explicit causal semantics from unstructured data effectively insulates the forecasting model against severe market non-stationarity, enabling robust performance even during unprecedented price spikes.

To qualitatively assess prediction performance, two case studies are presented along with their visualization results. Case 1 represents a short-term forecasting scenario (96-h history → 24-h prediction), whereas Case 2 represents a long-term scenario (96-h history → 72-h prediction). As illustrated in Fig. 9, in Case 1, the proposed RIS-LLM accurately captures the overall fluctuation patterns of electricity prices.

Table 4
Analyses of forecasting results on high-volatility periods.

Methods	MAE	MSE
XGBOOST [41]	0.781	0.899
CNN-LSTM [42]	0.365	0.413
Autoformer [43]	0.379	0.434
DLinear [44]	0.380	0.445
Time-LLM [26]	0.620	0.766
RIS-LLM	**0.273**	**0.307**

The predicted curve closely follows the ground truth, especially in regions with sharp upward or downward movements, demonstrating the model's ability to learn nonlinear volatility dynamics. Although Time-LLM, DLinear, and Autoformer also succeed in reproducing the general fluctuation trend, their predicted amplitudes are noticeably lower than the actual price variations. This underestimation mainly arises from their limited capability in distinguishing high-frequency volatility signals from trend components, causing smoothing effects around local extremes. DLinear and Autoformer both exhibit an underestimated extremum. Their reliance on linear decomposition and standard temporal attention inherently limits their capacity to capture high-frequency volatility signals, causing smoothing effects around local extremes. CNN-LSTM displays a significant trend deviation, completely missing the sharp downward trajectory and predicting a flat, misaligned curve. XGBoost shows the largest deviation in both fluctuation timing and amplitude, with poor alignment to the actual trend. The gradient-boosting structure is inherently less effective in capturing continuous temporal dependencies, resulting in discontinuous and delayed predictions. In Case 2, Fig. 10 further shows that RIS-LLM continues to demonstrate strong robustness under the more challenging long-horizon forecasting setting. The model successfully identifies several critical fluctuation segments, though slight overestimations appear in some rapid downward transitions. This behavior can be attributed to accumulated uncertainty over longer horizons. Time-LLM exhibits extreme valley misalignment, failing entirely to predict the deep drops, which indicates that directly patching raw time series struggles with long-range non-stationary shifts. DLinear and Autoformer capture the basic periodicity but suffer from insufficient trend intensity and an underestimated peak amplitude, respectively. This behavior is attributed to the accumulated uncertainty and inherent smoothing effects over extended horizons. CNN-LSTM displays an unstable prediction trajectory, severely distorting the fluctuation amplitudes over time. XGBoost

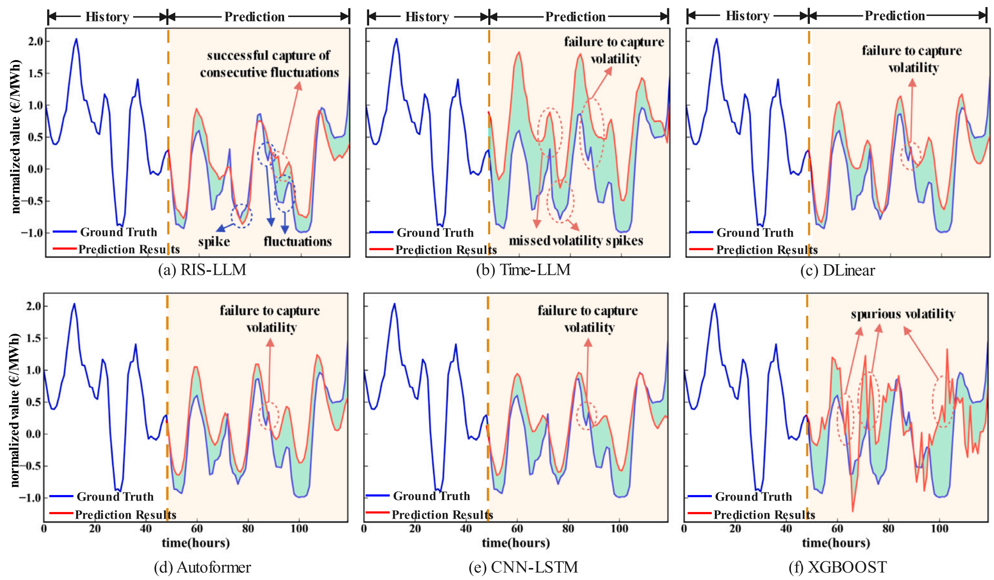

Fig. 8. Forecasting results on high-volatility periods.

almost collapses in the long-term forecasting task, showing highly noisy and unpredictable spikes, as its tree-based regression structure lacks the dynamic representation capacity required for long temporal dependencies.

Overall, both the quantitative metrics and qualitative visualizations confirm that the RIS-LLM framework achieves superior forecasting fidelity across both short- and long-term horizons. Crucially, the model demonstrates exceptional resilience under extreme market conditions, successfully capturing complex volatility spikes and deep valleys where conventional baselines suffer from severe smoothing effects or structural collapse. By integrating adaptive decomposition, causal semantics alignment, and reasoning-guided large model inference, the proposed approach effectively bridges the semantic gap between high-dimensional numerical data and physical market drivers. Instead of relying on purely data-driven statistical extrapolation, RIS-LLM provides a structured, causal representation of energy system states. This paradigm shift from implicit numerical curve-fitting to explicit semantic reasoning fundamentally differentiates the proposed framework from traditional forecasting models, underscoring its profound relevance and practical value for engineering informatics applications in modern, highly volatile electricity markets.

4.4. Ablation studies

To further assess the contributions of each key component in the RIS-LLM framework and substantiate the necessity of the LLM backbone, comprehensive ablation experiments were conducted. The selected scenario corresponds to long-horizon forecasting (96-hour history → 72-hour prediction), which presents the greatest challenge for capturing complex temporal dependencies and performing semantic reasoning.

The ablation and comparative study results are summarized in Table 5. The baseline model is the complete RIS-LLM, while the following variants are considered:

- **RFE + TSS + Autoformer**: To explicitly disentangle the contributions of semantic feature construction from LLM-based reasoning, a variant where the raw time series is pre-processed by our RFE and TSS modules before being fed into the Autoformer, bypassing the LLM.
- **w/o RFE**: Removes the RFE module, directly feeding raw time series into the semantic alignment stage. This tests the importance of multilevel decomposition and structured feature representation.

Table 5
Component ablation study results of RIS-LLM.

Variants	MAE	MSE
RIS-LLM	0.599	0.705
Autoformer	0.701	0.835
RFE + TSS + Autoformer	0.625	0.758
w/o RFE	0.613	0.752
w/o ALOESS	0.605	0.721
w/o TSS	0.648	0.769
w/o CM	0.609	0.744

- **w/o ALOESS**: Replaces the adaptive LOESS decomposition with a fixed smoothing parameter, evaluating the role of adaptive smoothing in separating trend and fluctuation components.
- **w/o TSS**: Disables the TSS alignment mechanism, feeding multivariate features into the LLM without semantic mapping, thereby assessing the contribution of semantic alignment.
- **w/o CM**: Removes the predictive dependency matrix from the attention network, effectively reducing causal-guided attention to standard self-attention.

From Table 5, several pivotal observations can be drawn. First, evaluating the necessity of the LLM, the integration of RFE and TSS significantly improves the standard Autoformer, which reduced its MAE from 0.701 to 0.625. However, the complete RIS-LLM framework achieves the lowest MAE of 0.599, outperforming the “feature-augmented” Autoformer. This confirms that while the RFE and TSS modules extract high-quality representations, the LLM-based reasoning provides measurable performance gains in translating extracted features into accurate forecasts. Regarding the internal components, the absence of the TSS module causes the most significant performance degradation. Without it, the MAE and MSE increase to 0.648 and 0.769, corresponding to 8.1% and 9.0% relative increases over the RIS-LLM baseline. This demonstrates that TSS plays a critical role in aligning temporal features and semantic reasoning across exogenous influencing factors. When the RFE module is removed, the MAE and MSE increase to 0.613 and 0.752, proving the necessity of structured feature representation. Eliminating the ALOESS mechanism also leads to a noticeable decrease in accuracy, confirming that adaptive smoothing is vital for precise trend decomposition under varying volatility levels. Finally, removing the predictive dependency matrix worsens

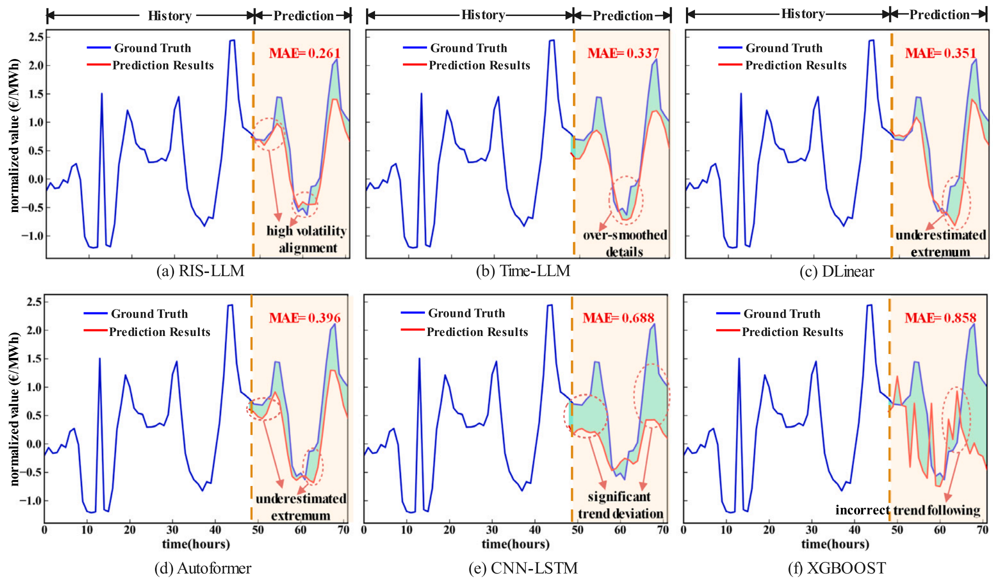

Fig. 9. Forecasting case under the horizon of 24.

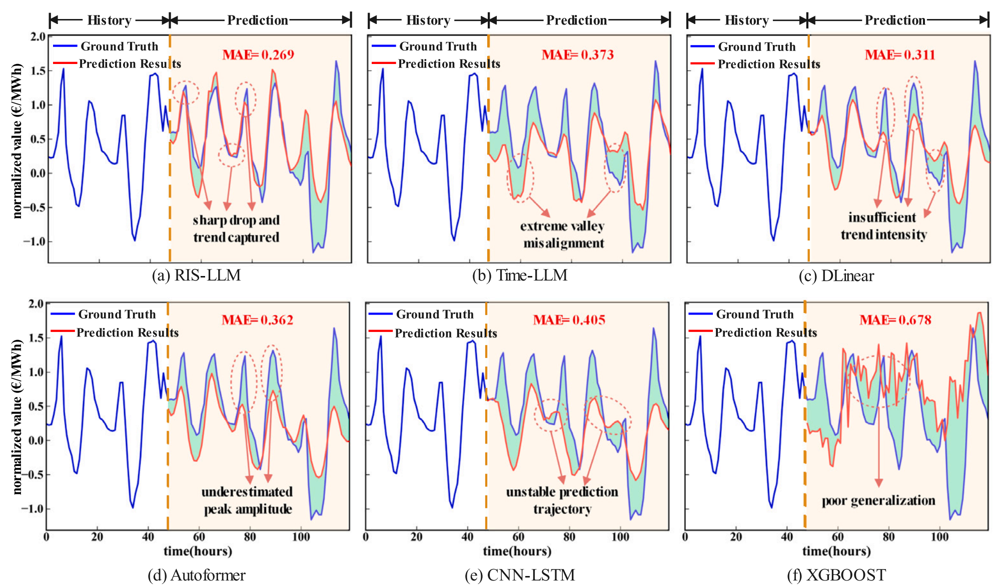

Fig. 10. Forecasting case under the horizon of 72.

Table 6
LLM backbones ablation study results of RIS-LLM.

Backbones	Params(M)	Model dimension	MAE	MSE
BERT-base	110	768	0.670	0.946
LLaMA3-1B	1000	2048	0.602	0.788
LLaMA3-3B	3000	3072	0.583	0.714
LLaMA-7B	7000	4096	**0.571**	**0.691**

performance by increasing the MAE to 0.609, indicating that causal-guided attention more effectively isolates physical market drivers than standard self-attention.

Overall, these ablation results confirm that each component in RIS-LLM contributes to improved predictive performance, with the TSS and RFE modules providing the largest gains.

To examine the influence of the reasoning core, an ablation study was conducted on various LLM backbones, including BERT-base, LLaMA3-1B, LLaMA3-3B, and LLaMA-7B. Their parameter counts, model dimension, and forecasting performance are summarized in Table 6.

The results show that BERT-base achieves relatively poor performance due to its limited contextual reasoning ability and lack of autoregressive prediction design. LLaMA3-1B and LLaMA3-3B exhibit progressive improvement with increasing model size, benefiting from deeper semantic encoding and stronger representation capacity. The complete RIS-LLM with LLaMA-7B achieves the best performance, reducing MAE and MSE by approximately 2%–3% compared with LLaMA3-3B. These findings indicate that larger reasoning models with enhanced contextual awareness are better able to integrate semantic constraints and temporal dependencies, yielding more accurate and stable electricity price forecasts.

5. Conclusion

This paper proposes the RIS-LLM framework, a reasoning-informed semantic framework for electricity market price dynamics modeling and forecasting. The core contribution of this work extends beyond merely achieving superior forecasting accuracy; it meaningfully advances the integration of semantic reasoning into energy forecasting. By integrating a RFE module and a TSS alignment mechanism, the proposed framework explicitly bridges the semantic gap between high-dimensional numerical sequences and actual physical market drivers. The results comprehensively demonstrate that the LLM extends beyond conventional sequence modeling by leveraging its pre-trained semantic knowledge to process structured feature representations, thereby providing measurable performance gains in electricity price forecasting. It successfully aligns complex exogenous events-such as sudden load surges or renewable oversupply-with resulting price dynamics, performing explicit semantic correlation reasoning that maintains exceptional structural robustness even during extreme market shifts where conventional baselines collapse.

Beyond its methodological contributions, the RIS-LLM framework may offer practical value for real-world electricity-market applications. For market participants, RIS-LLM can function as a decision-support tool for day-ahead and real-time bidding by providing both electricity-price forecasts and semantically interpretable representations of the factors associated with predicted market movements. Such contextual information may support more informed decision-making under volatile market conditions. In risk management, the framework's semantic characterization of high-volatility periods and price-spike events may contribute to scenario-aware monitoring and early-warning mechanisms. In addition, RIS-LLM may support improved contextual understanding of electricity-price dynamics associated with renewable-energy variability, thereby providing potentially useful insights for system operators in renewable-rich power systems. While these applications remain prospective directions requiring further real-world validation, they illustrate how LLM-enhanced semantic forecasting frameworks may contribute to future intelligent energy-market operations.

Despite these substantial advancements, the current framework exhibits certain practical limitations that warrant further investigation. First, from a modeling perspective, the semantic labeling process within the TSS module currently relies on unsupervised clustering. While this approach effectively captures macroscopic physical drivers, it may oversimplify the highly complex, multi-agent strategic gaming and bidding behaviors inherent in electricity markets. Second, from an engineering deployment perspective, although a relatively lightweight 7B open-source LLM was adopted to balance performance and efficiency, the computational overhead of autoregressive inference remains non-trivial. This latency presents a practical bottleneck for ultra-high-frequency trading applications or resource-constrained edge computing deployments in power grids.

Addressing these limitations paves the way for several highly promising directions for future research. One key area involves integrating multi-agent reinforcement learning with the RIS-LLM framework to evolve the model from a passive forecaster into an active decision-maker, capable of formulating LLM-driven, semantics-aware bidding strategies. Additionally, future studies should explore physics-informed lightweight model distillation techniques. By embedding rigorous physical grid constraints directly into more computationally efficient architectures, researchers can effectively close the loop between accurate semantic forecasting and real-time, optimal dispatch decision-making in next-generation energy systems.

CRediT authorship contribution statement

Ming Huang: Writing – original draft, Validation, Methodology. **Jieming Ma:** Writing – review & editing, Supervision, Conceptualization. **Yuxuan Luo:** Validation, Software. **Ka Lok Man:** Writing – review & editing, Supervision. **Sheng-Uei Guan:** Writing – review & editing, Supervision. **Xue Zhang:** Writing – review & editing, Supervision. **Xiaohui Zhu:** Writing – review & editing. **Eng Gee Lim:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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